A Project Report on

THE MOORE-PENROSE INVERSE

in the partial fulfillment of the requirements for the award of the degree of

M.Sc Applied Mathematics,

by

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May,2026

ACKNOWLEDGEMENT

I would like to take this opportunity to express my profound gratitude to my project supervisor Dr. Srita Ojha for her exemplary guidance and constant encouragement throughout the course of this project. I am grateful to IIEST, Shibpur for providing me the opportunity to complete my dissertation. In this project, I have tried to give a brief overview of area of The Moore-Penrose Inverse. I have benefited greatly from advantages of the LaTeX software typing system. It helped me a lot for writing my project easily.

I would like to extend my deepest gratitude and thanks to my parents without whose constant support and encouragement, this project could never have been completed. I would also like to thank my friends for their valuable suggestions and support for this project.

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1 Introduction

The classical inverse A^{-1} of a square matrix A exists if and only if A is non-singular. Yet in virtually every branch of applied mathematics—linear regression, signal processing, control theory, differential equations, and optimization—the matrices that arise are frequently rectangular, singular, or both. The question naturally emerges: is there a meaningful generalization of the matrix inverse to this broader class of matrices?

The answer, supplied independently by E.H. Moore (c. 1906–1920) and Roger Penrose (1955), is the Moore-Penrose generalized inverse, denoted $A^{\dagger}$. Moore called it the "general reciprocal"; Penrose characterized it through four elegant algebraic equations that uniquely determine $A^{\dagger}$ for any matrix A of any size and rank. When A is square and non-singular, $A^{\dagger} = A^{-1}$, so the Moore-Penrose inverse is a true generalization.

The significance of $A^{\dagger}$ is threefold. First, algebraically, it is the unique solution to the Penrose equations, placing it at the top of a rich hierarchy of generalized inverses. Second, geometrically and analytically, it produces the least-squares solution of minimum norm to any linear system, connecting it to fundamental problems in optimization. Third, computationally, it is accessible through several stable numerical algorithms, making it practically tractable for large-scale applications.

2 Existence and Construction of Generalized Inverses

2.1 The Penrose Equations

In 1955 Penrose showed that, for every finite matrix A (square or rectangular) of real or complex entries, there exists a unique matrix X satisfying the four equations

$$AXA = A \tag{1}$$
$$XAX = X \tag{2}$$
$$(AX)^* = AX \tag{3}$$
$$(XA)^* = XA \tag{4}$$

where A^* denotes the conjugate transpose of A. The unique matrix satisfying all four equations is called the Moore-Penrose inverse of A, denoted by $A^{\dagger}$. If A is non-singular, then $X = A^{-1}$ satisfies all four Penrose equations. Hence,

$$A^{\dagger} = A^{-1}$$

for every non-singular matrix A.

2.1.1 Definition of $\{i, j, \ldots, k\}$-Inverse

For any matrix $A \in \mathbb{C}^{m\times n}$, let A{i,j,. . . ,k} denote the set of matrices $X \in \mathbb{C}^{n\times m}$ satisfying equations $(i), (j), \ldots, (k)$ among the Penrose equations. A matrix $X \in A\{i, j, \ldots, k\}$ is called an {i,j,. . . ,k}-inverse of A. Some important examples are:

- $A\{1\}$: matrices satisfying AXA=A
- $A\{1, 2\}$: matrices satisfying AXA=A and XAX=X

- $A\{1,2,3,4\}$: the Moore-Penrose inverse

Uniqueness of the Moore–Penrose Inverse:

Example. If $A\{1,2,3,4\}$ is nonempty, then it consists of a single element.

Solution. Let $X, Y \in A\{1,2,3,4\}$ Then both X and Y satisfy the four Penrose equations. Using equations (2) and (3), $X = X(XA)^*$ Since $(XA)^* = XA$, we obtain $X = XX^*A^*$ Similarly, $X = X(AX)^*(AY)^*$ because $(AX)^* = AX$ and $(AY)^* = AY$ Hence, $X = XAY$ Using equation (4),

$$(XA)^* = XA, \qquad (YA)^* = YA$$

and simplifying, we obtain $X = Y$ Therefore, $A\{1,2,3,4\}$ contains exactly one element. Hence the Moore–Penrose inverse is unique.

$$\boxed{A^\dagger \text{ is unique}}$$

$A^\dagger$ is called the Moore-Penrose inverse

2.2 Existence and Construction of {1}-Inverses

It is easy to construct a $\{1\}$-inverse of the matrix $R \in \mathbb{C}_r^{m\times n}$ given by

$$R = \begin{bmatrix} I_r & K \\ O & O \end{bmatrix}. \tag{0.71}$$

For any matrix $L \in \mathbb{C}^{(n-r)\times(m-r)}$, the $n \times m$ matrix

$$S = \begin{bmatrix} I_r & O \\ O & L \end{bmatrix}$$

is a $\{1\}$-inverse of R. If R is of full column rank or full row rank, then the corresponding lower or right-hand sub-matrices are interpreted as absent.

2.2.1 Construction for an Arbitrary Matrix

The construction of $\{1\}$-inverses for an arbitrary matrix $A \in \mathbb{C}^{m\times n}$ is simplified by transforming A into a Hermite normal form.

Theorem 1. Let $A \in \mathbb{C}_r^{m\times n}$ and let $E \in \mathbb{C}^{m\times m}$, $P \in \mathbb{C}^{n\times n}$ be non-singular matrices such that

$$EAP = \begin{bmatrix} I_r & K \\ O & O \end{bmatrix}. \tag{1}$$

Then, for any $L \in \mathbb{C}^{(n-r)\times(m-r)}$, the $n \times m$ matrix

$$X = P \begin{bmatrix} I_r & O \\ O & L \end{bmatrix} E \tag{2}$$

is a $\{1\}$-inverse of A. The partitioned matrices in (1) and (2) must be suitably interpreted in the cases $r = m$ or $r = n$.

Proof.

Rewriting equation (1), we obtain

$$A = E^{-1} \begin{bmatrix} I_r & K \\ O & O \end{bmatrix} P^{-1}. \tag{3}$$

It is easily verified that any matrix X given by (2) satisfies $AXA = A$. Hence X is a $\{1\}$-inverse of A. In the trivial case $r = 0$, the matrix A is the $m \times n$ null matrix, and therefore any $n \times m$ matrix is a $\{1\}$-inverse.

2.2.2 Rank of a $\{1\}$-Inverse

Since P and E are non-singular, the rank of X equals the rank of the partitioned matrix appearing in equation (2). Therefore,

$$\operatorname{rank}(X) = r + \operatorname{rank}(L). \tag{4}$$

Since L is arbitrary, a $\{1\}$-inverse of A exists having any rank between

$$r \quad \text{and} \quad \min\{m, n\},$$

inclusive. Thus every finite matrix over the complex field possesses a $\{1\}$-inverse, and Theorem 1. provides a constructive method for obtaining such inverses.

2.3 Properties of $\{1\}$-Inverses

For a given matrix A, we denote any $\{1\}$-inverse by $A^{(1)}$. In general, $A^{(1)}$ is not uniquely defined. For any scalar λ, define

$$\lambda^{\dagger} = \begin{cases} \lambda^{-1}, & \text{if } \lambda \neq 0, \\ 0, & \text{if } \lambda = 0. \end{cases} \tag{1}$$

Lemma 1. Let $A \in \mathbb{C}_r^{m \times n}, \lambda \in \mathbb{C}$ Then:

(a) $\left(A^{(1)}\right)^* \in A^*\{1\}$

(b) If A is non-singular, then $A^{(1)} = A^{-1}$ uniquely.

(c) $\lambda^{\dagger} A^{(1)} \in (\lambda A)\{1\}$

(d) $\operatorname{rank}(A^{(1)}) \geq \operatorname{rank}(A)$

(e) If S and T are non-singular, then $T^{-1}A^{(1)}S^{-1} \in (SAT)\{1\}$

(f) The matrices $AA^{(1)}$ and $A^{(1)}A$ are idempotent and have the same rank as A.

Proof.

These results are immediate consequences of the defining relation $AXA = A$. Parts (d) and the latter part of (f) use the fact that the rank of a product of matrices does not exceed the rank of any factor. If an $m \times n$ matrix A has full column rank, then its $\{1\}$-inverses are precisely its left inverses. If A has full row rank, then its $\{1\}$-inverses are precisely its right inverses.

Lemma 2. Let $A \in \mathbb{C}_r^{m\times n}$. Then

(a) $A^{(1)}A = I_n$ if and only if $r = n$.

(b) $AA^{(1)} = I_m$ if and only if $r = m$.

Proof.

(a) If part: Let $A \in \mathbb{C}_r^{m\times n}$. Then the $n \times n$ matrix $A^{(1)}A$ is, by Lemma 1(f), idempotent and non-singular. Since

$$(A^{(1)}A)^2 = A^{(1)}A,$$

multiplying both sides by $(A^{(1)}A)^{-1}$ gives $A^{(1)}A = I_n$.

Only if part: If $A^{(1)}A = I_n$, then $\operatorname{rank}(A^{(1)}A) = n$. Hence, $\operatorname{rank}(A) = n$, by Lemma 1(f). Therefore, $r = n$.

(b) Similarly proved.

2.4 Existence and Construction of $\{1, 2\}$-Inverses

It was first noted by Bjerhammar that the existence of a $\{1\}$-inverse of a matrix A implies the existence of a $\{1, 2\}$-inverse.

Lemma 3. Let $Y, Z \in A\{1\}$, and let$X = YAZ$. Then$X \in A\{1, 2\}$.

Proof.

Since $Y, Z \in A\{1\}$,

$$AYA = A, \qquad AZA = A.$$

Now, $AXA = A(YAZ)A = (AYA)ZA = AZA = A$. Hence $X \in A\{1\}$. Also, $XAX = (YAZ)A(YAZ) = YA(ZAYA)Z$. Using $AZA = A$, $ZAYA = ZA$. Therefore, $XAX = YAZ = X$. Thus, $X \in A\{1, 2\}$. Since the matrices A and X occur symmetrically in the equations $AXA = A$ and $XAX = X$, the statements $X \in A\{1, 2\}$ and $A \in X\{1, 2\}$ are equivalent. Hence A and X are said to be $\{1, 2\}$-inverses of each other. From equations (1) and (2), together with the fact that the rank of a product of matrices cannot exceed the rank of any factor, it follows that if A and X are $\{1, 2\}$-inverses of each other, then

$$\operatorname{rank}(A) = \operatorname{rank}(X).$$

Therefore, if X is a $\{1\}$-inverse of A and has the same rank as A, then X is a $\{1, 2\}$-inverse of A.

Theorem 2 (Bjerhammar). Given A and $X \in A\{1\}$, $X \in A\{1, 2\}$ if and only if $\operatorname{rank}(X) = \operatorname{rank}(A)$.

Proof.

If part: Clearly, $R(XA) \subseteq R(X)$. But by Lemma 1(f), $\text{rank}(XA) = \text{rank}(A)$. Hence, if $\text{rank}(X) = \text{rank}(A)$, then $R(XA) = R(X)$. Therefore, $XAY = X$ for some matrix Y. Pre-multiplying by A,

$$AX = AXAY = AY.$$

Hence, $XAX = X$. Thus, $X \in A\{2\}$. Since $X \in A\{1\}$ already, we obtain $X \in A\{1, 2\}$.
Only if part: If $X \in A\{1, 2\}$, then $AXA = A$ and $XAX = X$. These immediately imply

$\text{rank}(X) = \text{rank}(A)$.

Corollary 1. Any two of the following three statements imply the third:

(i) $X \in A\{1\}$,

(ii) $X \in A\{2\}$,

(iii) $\text{rank}(X) = \text{rank}(A)$.

2.5 Explicit Formula for $A^\dagger$

Theorem 5 (MacDuffee). Let $A \in \mathbb{C}_r^{m\times n}$, $r > 0$, and suppose that A has a full-rank factorization

$$A = FG, \tag{16}$$

where $F \in \mathbb{C}^{m\times r}$, $G \in \mathbb{C}^{r\times n}$. Then

$$A^\dagger = G^*(F^*AG^*)^{-1}F^*. \tag{17}$$

Proof.

First, we show that the matrix F^*AG^* is nonsingular. Using the factorization $A = FG$,

$$F^*AG^* = F^*(FG)G^* = (F^*F)(GG^*) \tag{18}$$

Both factors on the right-hand side are $r \times r$ matrices. By Lemma 4,

$$\text{rank}(F^*F) = r, \qquad \text{rank}(GG^*) = r.$$

Hence, both matrices are non-singular. Therefore, F^*AG^* is the product of two nonsingular matrices, and so it is nonsingular. Moreover, from equation (18),

$$(F^*AG^*)^{-1} = (GG^*)^{-1}(F^*F)^{-1}$$

Let $X = G^*(F^*AG^*)^{-1}F^*$. Then,

$$X = G^*(GG^*)^{-1}(F^*F)^{-1}F^*. \tag{19}$$

It can be verified directly that the matrix X satisfies the four Penrose equations: $AXA = A$, $XAX = X$, $(AX)^* = AX$, and $(XA)^* = XA$. Hence, $X = A^\dagger$. Therefore, $A^\dagger = G^*(F^*AG^*)^{-1}F^*$.

2.6 Existence and Construction of $\{1,2,3\}$-, $\{1,2,4\}$-, and $\{1,2,3,4\}$-Inverses

Lemma 4. For any finite matrix A,

$$\operatorname{rank}(AA^*) = \operatorname{rank}(A) = \operatorname{rank}(A^*A). \tag{1}$$

Proof.

Let$A \in \mathbb{C}^{m\times n}$. Both A and AA^* have m rows. The rank of any m-rowed matrix equals m minus the number of independent linear relations among its rows. To prove

$$\operatorname{rank}(AA^*) = \operatorname{rank}(A),$$

it is sufficient to show that every linear relation among the rows of A holds for the corresponding rows of AA^*, and conversely. A nontrivial linear relation among the rows of a matrix H is equivalent to the existence of a nonzero row vector x^* such that $x^*H = 0$. Now,evidently $x^*A = 0 \quad \Rightarrow \quad x^*AA^* = 0$. and, conversely $x^*AA^* = 0$ implies

$$0 = x^*AA^*x = (A^*x)^*(A^*x).$$

Hence, $A^*x = 0$, which implies $x^*A = 0$. Here we use the fact that for any complex vector y, y^*y is the sum of squares of the absolute values of its entries, and this sum vanishes only when every entry is zero. Applying the same argument to the matrix A^*, we obtain $\operatorname{rank}(A^*A) = \operatorname{rank}(A^*)$. Since $\operatorname{rank}(A^*) = \operatorname{rank}(A)$, the result follows.

Corollary 2. For any finite matrix A,

$$R(AA^*) = R(A)$$

and

$$N(AA^*) = N(A).$$

Theorem 3 (Urquhart). For every finite matrix A with complex entries,

$$Y = (A^*A)^{(1)}A^* \in A\{1,2,3\}, \tag{12a}$$

and

$$Z = A^*(AA^*)^{(1)} \in A\{1,2,4\}. \tag{12b}$$

Proof.

Applying Corollary 2 to A^*, we obtain $R(A^*A) = R(A^*)$. Hence,

$$A^* = A^*AU \tag{13}$$

for some matrix U. Taking conjugate transpose of (13),

$$A = U^*A^*A. \tag{14}$$

Now, $AYA = U^*A^*A(A^*A)^{(1)}A^*A = U^*A^*A = A$. Therefore, $Y \in A\{1\}$. By Lemma 1(d), $\operatorname{rank}(Y) \geq \operatorname{rank}(A)$, while from the definition of Y, $\operatorname{rank}(Y) \leq \operatorname{rank}(A^*) = \operatorname{rank}(A)$. Hence, $\operatorname{rank}(Y) = \operatorname{rank}(A)$. By Theorem 2, $Y \in A\{1,2\}$. Finally, using (13) and (14),

$$AY = U^*A^*A(A^*A)^{(1)}A^*AU = U^*A^*AU.$$

Since U^*A^*AU is Hermitian, $(AY)^* = AY$. Therefore, $Y \in A\{1,2,3\}$. Thus, equation (12a) is established. The proof of (12b) is similar.

3 Minimal Properties of Generalized Inverses

3.1 Least-Squares Solutions of Inconsistent Linear Systems

Theorem 1. Let $A \in \mathbb{C}^{m\times n}$, $b \in \mathbb{C}^m$. Then $\|Ax - b\|$ is smallest when

$$x = A^{(1,3)}b,$$

where $A^{(1,3)} \in A\{1,3\}$. Conversely, if $X \in \mathbb{C}^{n\times m}$ has the property that, for all b, $\|Ax-b\|$ is smallest when $x = Xb$, then

$$X \in A\{1,3\}.$$

Proof.

We know,

$$b = (P_{R(A)} + P_{R(A)^\perp})b. \tag{4}$$

Therefore, $b - Ax = (P_{R(A)}b - Ax) + P_{N(A^*)}b$. Hence,

$$\|Ax - b\|^2 = \|Ax - P_{R(A)}b\|^2 + \|P_{N(A^*)}b\|^2, \tag{5}$$

Evidently, (5) assumes its minimum value if and only if

$$Ax = P_{R(A)}b, \tag{6}$$

which holds if $x = A^{(1,3)}b$ for any $A^{(1,3)} \in A\{1,3\}$, we know

$$AA^{(1,3)} = P_{R(A)}. \tag{7}$$

Conversely, if X is such that for all b, $\|Ax - b\|$ is smallest when $x = Xb$, then (6) gives $AXb = P_{R(A)}b$ for all b, and therefore $AX = P_{R(A)}$. Thus $X \in A\{1,3\}$.

Example.

Consider the inconsistent linear system $Ax = b$, where

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Since the two equations $x_1 + x_2 = 2$ and $x_1 + x_2 = 3$ cannot hold simultaneously, the system is inconsistent. The Moore-Penrose inverse of A is

$$A^\dagger = \frac{1}{4}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

By Theorem 1, the least-squares solution is $x = A^\dagger b$. Hence,

$$x = \frac{1}{4}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 5 \\ 5 \end{bmatrix} = \begin{bmatrix} \frac{5}{4} \\ \frac{5}{4} \end{bmatrix}.$$

Now,

$$Ax = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} \frac{5}{4} \\ \frac{5}{4} \end{bmatrix} = \begin{bmatrix} \frac{5}{2} \\ \frac{5}{2} \end{bmatrix}.$$

Thus,

$$Ax - b = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix},$$

and the residual norm is minimized. Therefore,

$$x = \begin{bmatrix} \frac{5}{4} \\ \frac{5}{4} \end{bmatrix}$$

is the least-squares solution of minimum residual.

Corollary 1. A vector x is a least-squares solution of $Ax = b$ if and only if $Ax = P_{R(A)}b = AA^{(1,3)}b$. Thus, the general least-squares solution is

$$x = A^{(1,3)}b + (I_n - A^{(1,3)}A)y, \tag{8}$$

with $A^{(1,3)} \in A\{1,3\}$ and arbitrary $y \in \mathbb{C}^n$.

Theorem 2. Let $A \in \mathbb{C}^{m\times n}$, $b \in \mathbb{C}^m$. If $Ax = b$ has a solution for x, the unique solution for which $\|x\|$ is smallest is given by

$$x = A^{(1,4)}b,$$

where $A^{(1,4)} \in A\{1,4\}$. Conversely, if $X \in \mathbb{C}^{n\times m}$ is such that, whenever $Ax = b$ has a solution, $x = Xb$ is the solution of minimum norm, then $X \in A\{1,4\}$.

Proof.

If $Ax = b$ is consistent, then for any $A^{(1,4)} \in A\{1,4\}$, $x = A^{(1,4)}b$ is a solution that lies in $R(A^*)$, and thus it is the unique solution in $R(A^*)$. Therefore, it is the unique minimum-norm solution. Conversely, let X be such that, for all $b \in R(A)$, $x = Xb$ is the minimum-norm solution of $Ax = b$. Setting b equal to each column of A, in turn, we conclude that $XA = A^{(1,4)}A$ and $X \in A\{1,4\}$

Example.

Consider the system $Ax = b$, where

$$A = \begin{bmatrix} 1 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 2 \end{bmatrix}.$$

The system becomes $x_1 + x_2 = 2$. This system has infinitely many solutions:

$$x = \begin{bmatrix} 2-t \\ t \end{bmatrix}, \qquad t \in \mathbb{R}.$$

The Moore-Penrose inverse of A is

$$A^\dagger = \frac{1}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

By Theorem 2, the minimum norm solution is $x = A^\dagger b$. Therefore,

$$x = \frac{1}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix}(2) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Now compare norms:
$$\left\| \begin{bmatrix} 2-t \\ t \end{bmatrix} \right\|^2 = (2-t)^2 + t^2.$$
This expression is minimized when $t = 1$, giving $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Hence, $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is the unique minimum norm solution of the system.

Corollary 2 (Penrose). Let $A \in \mathbb{C}^{m\times n}$, $b \in \mathbb{C}^m$. Then, among the least-squares solutions of $Ax = b$, $A^\dagger b$ is the one of minimum norm. Conversely, if $X \in \mathbb{C}^{n\times m}$ has the property that, for all b, Xb is the minimum-norm least-squares solution of $Ax = b$, then $X = A^\dagger$.

Proof.

By Corollary 1, the least-squares solutions of $Ax = b$ coincide with the solutions of
$$Ax = AA^{(1,3)}b. \tag{6}$$
Thus the minimum-norm least-squares solution of $Ax = b$ is the minimum-norm solution of (6). But, by Theorem 2, the latter is $x = A^{(1,4)}AA^{(1,3)}b = A^\dagger b$ A matrix X having the properties stated above must satisfy $Xb = A^\dagger b$ for all $b \in \mathbb{C}^m$, and therefore $X = A^\dagger$.

The minimum-norm least-squares solution, $x_0 = A^\dagger b$, also called the approximate solution of $Ax = b$, is characterized by the inequalities
$$\|Ax_0 - b\| \le \|Ax - b\|, \qquad \text{for all } x, \tag{22a}$$
and
$$\|x_0\| < \|x\| \tag{22b}$$
for any $x \neq x_0$ gives equality in (22a).

3.2 Tikhonov Regularization

Theorem 3. The function $f_{\alpha^2}(x)$ has a unique minimizer x_{α^2} given by
$$x_{\alpha^2} = (A^*A + \alpha^2 I)^{-1}A^*b, \tag{39}$$
whose norm $\|x_{\alpha^2}\|$ is a monotone decreasing function of α^2.

Proof.

The function (37) is a special case of (15) with
$$k = 2, \qquad A_1 = A, \qquad A_2 = \alpha I, \qquad b_1 = b, \qquad b_2 = 0.$$
Substituting these values into (16), we get
$$(A^*A + \alpha^2 I)x = A^*b. \tag{40}$$

This has the unique solution (39), since $(A^*A + \alpha^2 I)$ is non-singular. Using Lemma 1, we may write

$$b = Av + u, \qquad v \in R(A^*), \quad u \in N(A^*). \tag{41}$$

Substituting into (39),

$$x_{\alpha^2} = (A^*A + \alpha^2 I)^{-1} A^*Av. \tag{42}$$

Let $\{v_1, v_2, \ldots, v_r\}$ be an orthonormal basis of $R(A^*)$ consisting of eigenvectors of A^*A corresponding to nonzero eigenvalues:

$$A^*Av_j = \sigma_j^2 v_j, \qquad (\sigma_j > 0,\ j = 1, \ldots, r). \tag{43}$$

If

$$v = \sum_{j=1}^{r} \beta_j v_j,$$

then (42) gives

$$x_{\alpha^2} = \sum_{j=1}^{r} \frac{\sigma_j^2 \beta_j}{\sigma_j^2 + \alpha^2} v_j.$$

Hence,

$$\|x_{\alpha^2}\|^2 = \sum_{j=1}^{r} \left(\frac{\sigma_j^2}{\sigma_j^2 + \alpha^2} \right)^2 |\beta_j|^2,$$

which is a monotone decreasing function of α^2.

Exercise. Let $A \in \mathbb{C}_r^{m\times n}$, $b \in \mathbb{C}^m$, and let $0 < p < \sqrt{r}\,\|u\|$, where u is given by

$$b = Av + u, \qquad v \in R(A^*), \quad u \in N(A^*). \tag{1}$$

Show that the constrained minimization problem minimize $\|Ax - b\|$ subject to $\|x\| = p$ has the unique solution

$$x = (A^*A + \alpha^2 I)^{-1} A^*b, \tag{2}$$

where α is uniquely determined by

$$\|(A^*A + \alpha^2 I)^{-1} A^*b\| = p. \tag{3}$$

Solution.

Consider the constrained minimization problem $\min \|Ax - b\|^2$ subject to $\|x\|^2 = p^2$. Using the method of Lagrange multipliers, define

$$L(x, \alpha) = \|Ax - b\|^2 + \alpha^2(\|x\|^2 - p^2).$$

Expanding, $L(x, \alpha) = (Ax - b)^*(Ax - b) + \alpha^2(x^*x - p^2)$. Differentiating with respect to x and equating to zero,

$$2A^*(Ax - b) + 2\alpha^2 x = 0.$$

Hence, $A^*Ax - A^*b + \alpha^2 x = 0$, which gives $(A^*A + \alpha^2 I)x = A^*b$. Since $A^*A + \alpha^2 I$ is non-singular for $\alpha > 0$, $x = (A^*A + \alpha^2 I)^{-1} A^*b$. Now impose the constraint $\|x\| = p$. Therefore,

$$\|(A^*A + \alpha^2 I)^{-1} A^*b\| = p.$$

To prove uniqueness, define $f(\alpha) = \|(A^*A + \alpha^2 I)^{-1}A^*b\|$. Using the spectral decomposition of A^*A,

$$f(\alpha)^2 = \sum_{j=1}^{r} \left(\frac{\sigma_j^2}{\sigma_j^2 + \alpha^2} \right)^2 |\beta_j|^2,$$

where $\sigma_j > 0$ are singular values of A. Each term $\frac{\sigma_j^2}{\sigma_j^2+\alpha^2}$ is strictly decreasing in α. Hence $f(\alpha)$ is strictly decreasing and continuous. Moreover,

$$\lim_{\alpha \to 0} f(\alpha) = \sqrt{r}\,\|u\|, \qquad \lim_{\alpha \to \infty} f(\alpha) = 0.$$

Since $0 < p < \sqrt{r}\,\|u\|$, there exists a unique $\alpha > 0$ such that $f(\alpha) = p$. Therefore the constrained minimization problem has the unique solution

$$x = (A^*A + \alpha^2 I)^{-1}A^*b.$$

3.3 Least-Squares Solutions and Basic Solutions

Let $N(A) = \{(I, J) : A_{IJ}$ is a non-singular $r \times r$ sub-matrix of $A\}$, where I and J denote sets of row and column indices respectively.

Theorem 4 (Berg). Let $A \in \mathbb{R}_r^{m\times n}$, $b \in \mathbb{R}^m$. Then the minimum-norm least-squares solution of Ax=b is the convex combination

$$x = \sum_{(I,J)\in N(A)} \lambda_{IJ} \widehat{A_{IJ}^{-1}}\, b_I, \tag{89}$$

Proof.

This follows by substituting (79) into RHS(81). Then (89) follows from (82) with weights $\lambda_{IJ} = \mu_I^* \nu_J^*$, which, by (80) and (0.83), are given by (77). Since (89) holds for all b, we obtain Berg's representation

$$A^\dagger = \sum_{(I,J)\in N} \lambda_{IJ} \widehat{A_{IJ}^{-1}}, \tag{75}$$

where $\widehat{A_{IJ}^{-1}}$ is an $n\times m$ matrix with the inverse of A_{IJ} in position (J, I) and zeros elsewhere. Consider next the weighted least-squares problem

$$\text{minimize } \|D^{1/2}(Ax - b)\|, \tag{90}$$

where $D = \operatorname{diag}(d_i)$ is a diagonal matrix with all weights $d_i > 0$.

Conclusion

This report presented the Moore-Penrose Inverses the canonical generalized inverse satisfying the Penrose equations and providing minimum norm least- squares solutions of linear systems. Its algebraic structure,optimization properties, and computational representations through full-rank factorization and singular value decomposition establish its central importance in linear algebra and applied mathematics. The Moore-Penrose inverse continues to play a significant role in numerical analysis, optimization machine learning, and scientific computing.

References

1. Ben-Israel, A. and Greville, T.N.E. (2003). *Generalized Inverses: Theory and Applications*, 2nd edition. Springer, New York.

2. Penrose, R. (1955). "A generalized inverse for matrices." *Mathematical Proceedings of the Cambridge Philosophical Society.*

3. Moore, E.H. (1920). "On the reciprocal of the general algebraic matrix." *Bulletin of the American Mathematical Society.*